

LESSON 7.11

MULTIPLYING INTEGERS USING MODELS



LESSON INTRODUCTION

MA.6.NSO.4.2 Apply and extend previous understandings of operations with whole numbers to multiply and divide integers with procedural fluency.

LEARNING TARGETS

- I can multiply integers using integer chips, number lines, and patterns.

BENCHMARK COHERENCE

BUILDING ON

MA.5.NSO.2 Add, subtract, multiply, and divide multi-digit numbers.

WORKING TOWARD

MA.7.NSO.2.2 Add, subtract, multiply, and divide rational numbers with procedural fluency.

MA.7.NSO.2.3 Solve real-world problems involving any of the four operations with rational numbers.

ESSENTIAL QUESTIONS

- How do you use number lines to multiply integers?
- What pattern can you use to determine the sign of the product of integers?

LESSON NARRATIVE

In this lesson, students continue to multiply integers, and in addition to using integer chips, they will learn to model the multiplication on number lines. Students also discover patterns with the signs of integer products and reflect on which is their preferred method for multiplying as they build fluency.

COMPONENT SUMMARY

7.11.1 WARM-UP (~5 MIN.)

Which One Doesn't Belong?

7.11.3 GUIDED PRACTICE (10 MIN.)

Multiply integers using integer chips and number lines

7.11.5 WRAP-UP (~10 MIN.)

Demonstrate understanding of lesson learning targets

7.11.2 GUIDED INSTRUCTION (10 MIN.)

Discover how to use number lines to model multiplying integers

7.11.4 YOUR TURN! (10 MIN.)

Practice multiplying integers using your preferred method

RESOURCES

GLOSSARY

Key Terms:

- N/A

Additional Terms:

- N/A

MATERIALS

- Number lines
- Integer chips

ADDITIONAL PREPARATION

- For 7.11.4 Your Turn! and 7.11.5 Wrap-Up, consider having prelabeled blank number lines available.

TEACHER TIPS

COMMON MISCONCEPTIONS

- Students may not recognize a pattern in the sign of the products when multiplying integers.
- Students may struggle to understand the concept of multiplying two negative integers.

THINGS TO CONSIDER

- Consider displaying the anchor chart from the Lesson 10 Wrap-Up to assist students in recognizing the patterns in the sign of the products of integers.
- Model the precise language of multiplication and how to model it on the number lines accurately in the 7.11.2 Guided Instruction.

LITERACY AND LANGUAGE ROUTINES

Which One Doesn't Belong

The 7.11.1 Warm-Up offers a low-floor, high-ceiling opportunity to engage with number lines. This instructional routine allows for a low barrier of entry, as any informal observations (mathematical or not) contribute to the discussion, while students can extend their thinking with a higher ceiling of possible mathematical answers for all four choices.

MATHEMATICAL THINKING AND REASONING (MTR) STANDARD

MA.K12.MTR.3.1 Mathematical Fluency

In 7.11.3 Guided Practice, students represent integer multiplication using both integer chips and number lines. Students progress from representing integer multiplication concretely to abstractly as they recognize and internalize the patterns with the signs of the products.

DIFFERENTIATE INSTRUCTION

Consider providing Study Expert videos or visual scaffolding from Lesson 10 throughout the lesson. Students should activate their prior knowledge while extending concepts of integer chips to number lines while building fluency with flexibility in problem-solving.

- 7.10.4 Exploration: Reference this activity or video from Lesson 10 for expressions with two opposite signs (-2×5).
- 7.10.6 Exploration: Provide this activity or video for multiplying a negative integer by a negative integer (like question 1b in 7.11.3).

7.11.1 WARM-UP: WHICH ONE DOESN'T BELONG?

Component Overview: Students revisit number-line models for adding integers to prepare for multiplying integers on number lines in the lesson. After examining each model, students determine which one they think doesn't belong with the others and justify with mathematics.

TEACHER MOVES

Which One Doesn't Belong

This instructional routine provides a low-floor, high-ceiling approach to engaging with number lines. Encourage students to choose more than one answer. After they have selected their answer choice and written an explanation, prompt them to find another answer with an explanation of why that one does not belong.

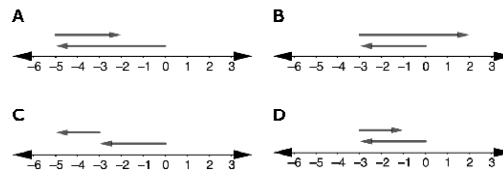
DIFFERENTIATE INSTRUCTION

If students struggle, consider providing students with sentence stems such as "I think the _____ doesn't belong because..."



7.11.1 Warm-Up: Which One Doesn't Belong?

1. Four number lines modeling addition are shown.



Explain which one doesn't belong.

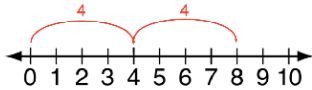
Answers will vary. C doesn't belong because both arrows are pointing to the left. Every other number line has one arrow starting at 0 and going left, showing a negative number, and one arrow going right, showing adding a positive number. C shows adding two negative numbers.



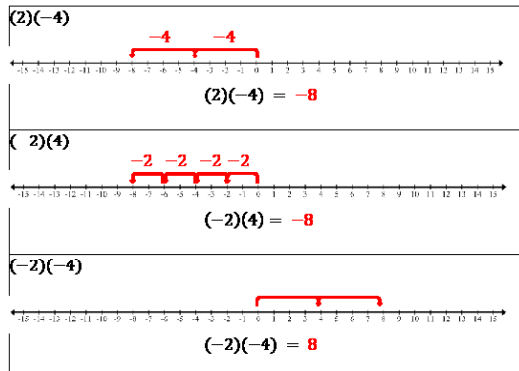
7.11.2 Guided Instruction: Multiplying Integers Using the Number Line

Recall that $(2)(4)$ means 2 groups with 4 units in each group. The product is the total number of objects.

This can be represented on the number line as shown.



1. Represent each multiplication expression on the number line. Then, use it to complete the equation.



2. Verify that the rules for multiplication you created using the integer chips match the results of multiplying with the number line. Make revisions to your rules if needed.

7.11.2 GUIDED INSTRUCTION: MULTIPLYING INTEGERS USING THE NUMBER LINE

Component Overview: Students are guided through how to model multiplying integers on a number line. Consider having students compare the use of number lines for multiplying integers to using integer chips.

DIFFERENTIATE INSTRUCTION

Consider displaying the examples, instructions, and visual models from this activity on an anchor chart.

TEACHER MOVES

Consider thinking aloud through each of the examples as you demonstrate how to model each expression on the number lines. This provides the opportunity to model precise language and ensure that each expression is modeled correctly on the number line. Consider using the following language for each of the expressions to model what students saw in Lesson 10:

$(2)(-4)$ “Two groups of negative 4.”

$(-2)(4)$ “The opposite of two groups of 4.”

$(-2)(-4)$ “The opposite of two groups of negative 4.”

This should be guided instruction, as it will be released to the students in the next component.

7.11.3 GUIDED PRACTICE: MULTIPLYING INTEGERS USING MODELS

Component Overview: Students model integer multiplication with integer chips and number lines. Students reflect on which representation they prefer to use and why.

TEACHER MOVES

Notice both representations of using integer chips for multiplying integers, with the second representation displaying how students used integer chips in Lesson 10.

Prompt students to think creatively and extend their thinking from Lesson 10 to include a different way to represent using integer chips.

Representation 1: “The opposite of two groups of negative two.”

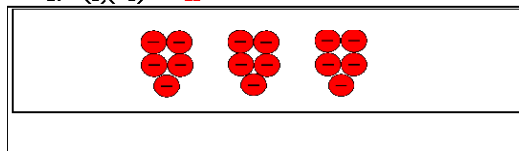
Representation 2: “Two groups of negative two should be removed.”



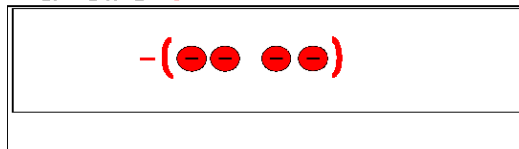
7.11.3 Guided Practice: Multiplying Integers Using Models

1. Use integer chips to model and evaluate each expression.

a. $(3)(-5) = -15$

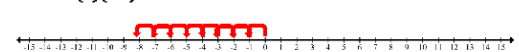


b. $-2 \times -2 = 4$



2. Use a number line model to evaluate each expression.

a. $(8)(-1) = -8$



b. $(-4)(-4) = 16$



3. Which model for multiplication of integers do you prefer: multiplication chips or the number line?
Answers will vary. Number line is faster.



7.11.4 Your Turn!

1. Find the value of each expression using your preferred method.

- a. 2×-8
-16

b. $-6 \cdot (-3)$
18

c. $(-7)(-6)$
42

d. $5(-3)$
-15

e. -9×-4
36

f. $7 \cdot 5$
35

7.11.4 YOUR TURN!

Component Overview: Students practice multiplying integers. Instructions include students “using your preferred method” to purposefully highlight and build fluency.

DIFFERENTIATE INSTRUCTION

Students develop agency over their learning by having the opportunity to choose which representation they prefer to use.

ENRICHMENT

- How did you solve the problem?
- Why did you choose this method?
- Can you solve it using another representation?



7.11.5 Wrap-Up: Ticket Out the Door

1. Find the value of each expression using your preferred method.

Expression	Model	Value
$-3 \cdot -3$		9
-2×4		-8
$(8)(-2)$		-16
$(3)(-8)$		-24

7.11.5 WRAP-UP: TICKET OUT THE DOOR

Component Overview: Students demonstrate understanding of the lesson learning target. Consider using the data from this formative assessment to determine which students need remediation before the next lesson. The table in this Wrap-Up could be used to create an anchor chart for future reference. Students should not use a calculator to complete the Wrap-Up.

DIFFERENTIATE INSTRUCTION

Provide students with integer chips or blank number lines if they struggle creating their own.

QUESTIONING

Challenge students to represent one expression using integer chips in two different ways. See 7.11.3 Guided Practice for more details.